

# Statistics II: Introduction to Inference

## Module 4: Concentration Inequalities and Sample Size Determination

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### 1 Some Inequalities

**(I) Markov's inequality:** Let  $X$  be a non-negative random variable. Then,

$$P(X \geq t) \leq \frac{\mathbb{E}(X)}{t}, \quad \text{for any } t > 0.$$

*Proof.* When the distribution of  $X$  has a pdf  $f_X$  then

$$\begin{aligned} \mathbb{E}(X) &= \int_0^\infty x f_X(x) dx = \int_0^t x f_X(x) dx + \int_t^\infty x f_X(x) dx \\ &\geq t \int_t^\infty f_X(x) dx = tP(X \geq t). \end{aligned}$$

Hence the proof.

For general random variable, we may extend the proof of Markov inequality as follows:

$$\mathbb{E}(X) = \mathbb{E}(X\mathbb{I}(X < t)) + \mathbb{E}(X\mathbb{I}(X \geq t)) \geq \mathbb{E}(X\mathbb{I}(X \geq t)) \geq t\mathbb{E}(\mathbb{I}(X \geq t)) = tP(X \geq t).$$

The last inequality follows from the fact that the random variable  $X\mathbb{I}(X \geq t) \geq t\mathbb{I}(X \geq t)$  with probability *one*, and expectation is monotone. This completes the proof.  $\square$

#### Applications of Markov's inequality:

- (1) **Chebyshev's inequality:** Let  $X$  be any random variable with expectation  $\mathbb{E}(X) = \mu$  and finite variance  $\text{Var}(X) = \sigma^2$ . Then,

$$P(|X - \mu| \geq \epsilon) = P(|X - \mu|^2 \geq \epsilon^2) \leq \frac{\mathbb{E}(|X - \mu|^2)}{\epsilon^2} = \frac{\sigma^2}{\epsilon^2} \quad (1)$$

by Markov's inequality. The above inequality is called the Chebyshev's inequality.

- (2) One can further sharpen the upper bound of  $P(|X - \mu| \geq \epsilon)$  (the tail probability) when higher order moments are finite.

Let  $X$  be a RV with expectation  $\mu$ , and finite  $r$ -th order moment  $\mu_r = \mathbb{E}|X - \mu|^r$ . Then,

$$P(|X - \mu| \geq \epsilon) \leq \frac{\mu_r}{\epsilon^r}, \quad r = 1, 2, \dots$$

**Note:** Usually one obtains a sharper bound by taking a larger value of  $r$ . However, for a larger value of  $r$ ,  $\mu_r$  may not be finite. Note that, for  $r > s$ , if  $\mu_r$  is finite then  $\mu_s$  is also finite, but the converse is NOT true. For example, for  $t_{(4)}$  distribution,  $\mu_2 = 2$ , but  $\mu_4 = \infty$ .

**(II) Cauchy–Schwarz inequality:** Let  $X$  and  $Y$  be two random variables such that  $\mathbb{E}(X), \mathbb{E}(Y)$  and  $\mathbb{E}(XY)$  exists, then

$$\{\mathbb{E}(XY)\}^2 \leq \mathbb{E}(X^2) \mathbb{E}(Y^2).$$

*Proof.* Define  $W = X - bY$  for some  $b \in \mathbb{R}$ . Observe that

$$\mathbb{E}(W^2) = \mathbb{E}[(X - bY)^2] = \mathbb{E}(X^2) + b^2\mathbb{E}(Y^2) - 2b\mathbb{E}(XY) \geq 0, \quad (2)$$

for any  $b \in \mathbb{R}$ . Choose  $b = \mathbb{E}(XY)/\mathbb{E}(Y^2)$ . Then the RHS of (2) becomes

$$\mathbb{E}(X^2) - \frac{\{\mathbb{E}(XY)\}^2}{\mathbb{E}(Y^2)} \geq 0,$$

which proves the inequality. Equality holds if and only if  $W = 0$  with probability (w.p.) 1, i.e.,  $X = bY$  w.p. 1 for some  $b \in \mathbb{R}$ .  $\square$

**(III) Application of the inequalities in minimum sample size determination:** A good estimator  $T_n$  of  $g(\theta)$  usually concentrates more around  $g(\theta)$ , as the sample size becomes larger. For example, the sample mean  $\bar{X}_n$  under a population with *(mean, variance)* =  $(\mu, \sigma^2)$  has expectation  $\mu$  and variance  $\sigma^2/n$ , which decreases as  $n$  increases. It follows that the probability distribution of  $\bar{X}_n$  will be more concentrated around the mean value  $\mu$  as  $n$  increases.

Now, suppose we want enlarge the sample size  $n$  so that we can ensure that with at least  $100(1-\alpha)\%$  probability,  $T_n$  concentrates around  $[g(\theta) - \epsilon, g(\theta) + \epsilon]$ , for some choice of  $\alpha \in (0, 1)$  and  $\epsilon > 0$ . Towards that, an appropriate probability bound turns out to be helpful in determining a minimum sample size which ensures such concentration. For example, suppose in the above setup we want to ensure 75% concentration of sample mean  $\bar{X}_n$  in  $[\mu - 0.3, \mu + 0.3]$ . In other words, we want a (lowest possible)  $n$  such that  $P(|\bar{X}_n - \mu| \leq 0.3) \geq 0.75$ . Suppose further that  $\sigma^2 = 2$ . Observe that, by Chebyshev's inequality

$$P(|\bar{X}_n - \mu| \leq 0.3) \geq 1 - \frac{2}{n \times 0.3^2}.$$

Thus, it is enough to choose  $n$  such that

$$1 - \frac{2}{n \times 0.3^2} \geq \frac{3}{4} \quad \implies \quad \frac{1}{4} \geq \frac{2}{n \times 0.3^2} \quad \implies \quad n \geq \frac{8}{0.3^2} = 88.89.$$

Therefore the minimum possible choice of  $n$  is 89 (as  $n$  must be a positive integer).

**Remark:** The inequality can be sharpened when additional information is available. Suppose it is further known that  $E[(\bar{X}_n - \mu)^4] = 3\sigma^4/n^2 = 12/n^2$  (which happens when the population is normally distributed), then we can apply the generalization of the Chebyshev's inequality (see (2)) with  $r = 4$ , and get a sharper bound of  $P(|\bar{X}_n - \mu| \leq 0.3)$ . By this inequality we have

$$P(|\bar{X}_n - \mu| \leq 0.3) \geq 1 - \frac{12}{n^2 \times 0.3^4}.$$

As before this can be ensured by choosing an  $n$  such that

$$1 - \frac{12}{n^2 \times 0.3^4} \geq \frac{3}{4} \quad \implies \quad n \geq \sqrt{\frac{48}{0.3^4}} = 76.98.$$

Therefore the minimum possible choice of  $n$  is 77.

**Another example [Bernoulli trials].** Suppose a fair coin is tossed independently  $n$  times. For  $i = 1, \dots, n$ , define

$$X_i = \begin{cases} 1, & \text{if a head occurs on the } i\text{th toss,} \\ 0, & \text{if a tail occurs on the } i\text{th toss.} \end{cases}$$

Then the sample mean  $\bar{X}_n$  represents the proportion of heads obtained in the  $n$  tosses. We want to determine how large  $n$  must be so that

$$P(0.4 \leq \bar{X}_n \leq 0.6) \geq 0.7.$$

Let  $T_n = \sum_{i=1}^n X_i$  denote the total number of heads observed in the  $n$  tosses. Then  $T_n$  follows a binomial distribution with parameters  $n$  and  $p = \frac{1}{2}$ . Consequently,

$$E(T_n) = \frac{n}{2}, \quad \text{Var}(T_n) = \frac{n}{4}.$$

Since  $\bar{X}_n = T_n/n$ , we obtain from Chebyshev's inequality

$$P(0.4 \leq \bar{X}_n \leq 0.6) = P(0.4n \leq T_n \leq 0.6n) = P\left(\left|T_n - \frac{n}{2}\right| \leq 0.1n\right).$$

Applying Chebyshev's inequality gives

$$P\left(\left|T_n - \frac{n}{2}\right| \leq 0.1n\right) \geq 1 - \frac{\text{Var}(T_n)}{(0.1n)^2} = 1 - \frac{n/4}{(0.1n)^2} = 1 - \frac{25}{n}.$$

In order to ensure 70% concentration, it is enough to choose  $n$  such that

$$1 - \frac{25}{n} \geq \frac{7}{10} \quad \implies \quad n \geq \frac{250}{3} = 83.3.$$

So, enough to take  $n = 84$ , this probability is at least 0.7, as required.